

the Poisson equation $-\nabla^2 u = f$,
and the Sobolev spaces $H^k(\Omega)$
calculations for weeks 12 & 13

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UAF Math 617 Functional Analysis

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Outline

- 1 the Poisson equation: $-\nabla^2 u = f$
- 2 the Poisson equation in weak form: $\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx$
- 3 the Sobolev spaces $H^k(\Omega)$
- 4 why does a Poisson solution exist?
- 5 why is that solution smooth?
- 6 lecture content in weeks 12 & 13

the Poisson equation

- let $\Omega \subset \mathbb{R}^d$ be open
- assume $d \in \{1, 2, 3\} \dots$ but I will usually draw pictures for $d = 2$
- assume the boundary of Ω is not too crazy (more below)
- assume $f = f(x) \in L^2(\Omega)$ is (square-)integrable

Poisson equation

find $u = u(x)$ so that

$$-\nabla^2 u = f \quad \text{on } \Omega$$

- next slide defines “ ∇^2 ” ...
- picture:

the Laplacian

- given f , finding u from the Poisson equation $-\nabla^2 u = f$ is a problem of inverting the operator $-\nabla^2$

Definition (Laplacian operator)

for $\Omega \subset \mathbb{R}^d$ open and $u : \Omega \rightarrow \mathbb{R}$, the *Laplacian* of u is

$$\nabla^2 u = \nabla \cdot (\nabla u) = \frac{\partial^2 u}{\partial x_1^2} + \cdots + \frac{\partial^2 u}{\partial x_d^2}$$

- $\nabla^2 u \stackrel{d=1}{=} u''$, $\nabla^2 u \stackrel{d=2}{=} u_{xx} + u_{yy}$, $\nabla^2 u \stackrel{d=3}{=} u_{xx} + u_{yy} + u_{zz}$

Definition (Laplace's equation = potential equation)

$$\nabla^2 u = 0,$$

and a solution of this equation is called *harmonic*

- $\nabla^2 = \nabla \cdot \nabla$ usually appears in models because a flux is proportional to a gradient, and then a divergence connects interior behavior to a boundary integral of the flux
- we will need the divergence theorem from the [weeks 10&11 slides](#):

$$\int_{\Omega} \nabla \cdot \mathbf{V} \, dm = \int_{\partial\Omega} \mathbf{V} \cdot \mathbf{n} \, ds,$$

- there is an $\mathbb{R}^d$ version of this theorem for any d
- when using this, $\mathbf{V}$ will be the gradient of a scalar function
- recall the boundary integral is the total flux of $\mathbf{V}$ outward through the boundary

boundary-value problems for the Poisson equation

- in the absence of any boundary conditions there is an infinite-dimensional space of solutions to $-\nabla^2 u = f$ on Ω
- if $f = 0$ then this is the space of all *harmonic functions* on Ω , an ∞ -dimensional subspace of all smooth functions:

$$\{w : \nabla^2 w = 0\} \subset C^2(\Omega)$$

- complex analysis is, essentially, the study of this space
- but we want to build models which make *one* prediction, so we will focus on *boundary-value problems* that pick-out one solution u

Dirichlet problem for the Poisson equation

for $f \in L^2(\Omega)$ and $g_D \in L^2(\partial\Omega)$, find $u : \Omega \rightarrow \mathbb{R}$ so that

$$\begin{aligned} -\nabla^2 u &= f & \text{on } \Omega \\ u &= g_D & \text{on } \partial\Omega \end{aligned}$$

boundary-value problems for the Poisson equation

- instead of u itself, the gradient ∇u could be set to known values along the boundary, which makes sense because the gradient is often proportional to a physical flux

Neumann problem for the Poisson equation

for $f \in L^2(\Omega)$ and $g_N \in L^2(\partial\Omega)$, find $u : \Omega \rightarrow \mathbb{R}$ so that

$$\begin{aligned} -\nabla^2 u &= f && \text{on } \Omega \\ \nabla u \cdot \mathbf{n} &= g_N && \text{on } \partial\Omega \end{aligned}$$

where $\mathbf{n}$ is the outward unit normal vector field along $\partial\Omega$

- other notation for the (scalar) directional derivative:

$$\nabla u \cdot \mathbf{n} = \frac{\partial u}{\partial n} = u_n$$

- one could have Dirichlet conditions on part of the boundary and Neumann on another part, so that the boundary is a disjoint union:

$$\partial\Omega = \partial_D\Omega \cup \partial_N\Omega$$

applications

- the Poisson equation can be used to model heat conduction, electrostatic potential, elastic membranes, equilibrium distributions for random walks, or many other physical phenomena

application: temperature of a conductive solid

- Fourier's law for heat conduction in solids says that the heat flux is $\mathbf{q} = -k\nabla u$, where u is the temperature, $\mathbf{q}$ is the heat flux, and $k > 0$ is the conductivity
- consider a solid with a heat capacity $c > 0$ (heat gained by given increase in temperature) and a mass density $\rho > 0$ (mass per unit volume)
- given a heat source f within the domain Ω , conservation of energy says

$$c\rho \frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{q} + f \quad \Longleftrightarrow \quad \frac{d}{dt} \left(\int_{\Omega} c\rho u \, dx \right) = - \int_{\partial\Omega} \mathbf{q} \cdot \mathbf{n} \, ds + \int_{\Omega} f \, dx$$

- assume steady state, so $\partial/\partial t = d/dt = 0$
- if k is constant then $-\nabla \cdot \mathbf{q} = -\nabla \cdot (-k\nabla u) = k\nabla^2 u$
- we derive Poisson's equation, the (steady) *heat* or *diffusion equation*

$$0 = k\nabla^2 u + f \quad \text{within } \Omega$$

application: deflection of an elastic membrane

- suppose $z = u(x, y)$ is a vertical displacement of an elastic membrane over a domain $\Omega \subset \mathbb{R}^2$
- suppose that along $\partial\Omega$ we have put a rigid frame with height $z = g(x, y)$
- suppose that the slopes along the graph of g are small
- suppose that the membrane is loaded by a force f (per unit area)
- group the elastic properties of the membrane into a constant $\kappa > 0$
- then the vertical displacement satisfies Poisson's equation

$$-\kappa \nabla^2 u = f \quad \text{within } \Omega$$

subject to $u = g$ along $\partial\Omega$

applications: Poisson equation boundary value problems

pictures for $d = 1$ steady heat in a rod, with insulated ends:

pictures for $d = 2$ membrane over a frame:

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6. lecture content in weeks 12 & 13

what is a “weak form”?

- FIXME strong form is pointwise
- weak form is averaged by a test function

picture of where the equation is enforced:

deriving the weak form of the Poisson equation

Fix u multiply by v and integrate by parts

$$\int_{\Omega} \nabla u \cdot \nabla v \, dx = \int_{\Omega} f v \, dx$$

- x

- FIXME Neumann through boundary integral
- FIXME Dirichlet through mod of vector space

what is a finite element method

FIXME localized v based on a mesh

FIXME $V_h \subset V$ is finite-dimensional

- x

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first Sobolev space

- given a bounded open set $\Omega \subset \mathbb{R}^d$
- recall the definition of (real-valued) $L^2(\Omega)$
- recall that we say f has a weak (partial) derivative $g = \partial f / \partial x_k$ if $g \in L^1(\Omega)$ so that

$$\int_{\Omega} g \varphi \, dm = - \int_{\Omega} f \frac{\partial \varphi}{\partial x_k} \, dm \quad \text{for all } \varphi \in C_c^\infty(\Omega)$$

- note that if Ω is bounded then $L^2(\Omega) \subset L^1(\Omega)$

Definition (Sobolev space $H^1(\Omega)$)

$$H^1(\Omega) = \left\{ f \in L^2(\Omega) \mid \begin{array}{l} \text{the weak partial derivatives } \frac{\partial f}{\partial x_k} \text{ exist} \\ \text{and are in } L^2(\Omega) \text{ for } k = 1, \dots, d \end{array} \right\}$$

- “ H ” is for Hilbert

more Sobolev spaces

- recall our multi-index derivative notation from the [weeks 10&11 slides](#)

Definition (weak derivative)

if $f \in L^1_{\text{loc}}(\Omega)$, and if $g \in L^1_{\text{loc}}(\Omega)$ satisfies

$$\int_{\Omega} g \varphi \, dm = (-1)^{|\alpha|} \int_{\Omega} f D^{\alpha} \varphi \, dm \quad \text{for all } \varphi \in C_c^{\infty}(\Omega)$$

then we say g is the α th weak derivative of f , and we write $D^{\alpha} f = g$ a.e.

Definition (Sobolev space $H^k(\Omega)$)

$$H^k(\Omega) = \left\{ f \in L^2(\Omega) \mid \begin{array}{l} \text{the weak derivatives } D^{\alpha} f \text{ exist} \\ \text{and are in } L^2(\Omega) \text{ for all } |\alpha| \leq k \end{array} \right\}$$

- we may write $H^0(\Omega) = L^2(\Omega)$
- other books use notation $H^k(\Omega) = W^{k,2}(\Omega)$; the “2” indicates that the weak derivatives are in $L^2(\Omega)$
- $W^{k,p}(\Omega)$ is the general case, with weak derivatives in $L^p(\Omega)$ up to order k

FIXME inner product and norm

FIXME follow Braess II.1 presentation (pages 28-31) re completion of

$S = H^k(\Omega) \cap C^\infty(\Omega)$ in H^k norm

- x

FIXME $H_0^k(\Omega)$ is completion of $C_c(\infty)$ in H^k norm

• X

FIXME Poincare-Friedrichs

• X

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FIXME: minimization route

• x

FIXME: Riesz representation theory route

• X

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FIXME: sketch how ellipticity means $u \in H^{2+k}(\Omega)$ if $f \in H^k(\Omega)$

• x

FIXME Fourier series along boundary converts to smooth in the interior
FIXME relate to definition of $H^k([-\pi, \pi])$

- x

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- most of this material is only from these slides

to know from slides:

- x

to prove:

- x
- Assignment 7 is posted at bueler.github.io/fa